

Automated analysis of embryonic gene expression with cellular resolution in *C. elegans*

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We describe a system that permits the automated analysis of reporter gene expression in *Caenorhabditis elegans* with cellular resolution continuously during embryogenesis. We demonstrate its utility by defining the expression patterns of reporters for several embryonically expressed transcription factors. The invariant cell lineage permits the automated alignment of multiple expression profiles, allowing direct comparison of the expression of different genes' reporters. We also used this system to monitor perturbations to normal development involving changes both in cell-division timing and in cell fate. Systematic application of this system could reveal the gene activity of each cell throughout development.

A major goal of current biological research is to understand how the genome directs the process by which a single-cell zygote gives rise to the complexity of a multicellular organism and in turn a new zygote. Simply knowing the full complement of transcriptionally active genes for each cell throughout development would be a major advance and would provide the molecular framework for understanding the network of interactions with which development proceeds. For example, the description of the expression pattern of pair-rule genes such as even skipped in *Drosophila melanogaster* with high spatial and temporal resolution had been critical in understanding how these genes can cause expressing cells to adopt a fate different from adjacent non-reporter-expressing cells¹.

Several different methods have attempted to capture expression information with high temporal and spatial resolution on a broad scale. Microarray experiments have provided valuable information such as temporal expression in whole embryos or single-time-point expression for specific cell types²⁻⁷. However, microarray-based analysis is unwieldy and technically challenging when trying to achieve both cellular and high temporal resolution. Imaging an organism more readily provides single-cell resolution of expression by using a transgenic reporter with a visualizable output, or by directly labeling transcripts or proteins. However, typically only one or a few genes are assayed at once, often only at specific times. Integrating such datasets across multiple genes and multiple

specimens throughout development generally requires expert anatomists, and expression profiles of individual cells are often lost.

The nematode *C. elegans* presents the possibility of new approaches to the comprehensive description of expression patterns. Because it is transparent from the zygote to the adult and has just 959 somatic cells as an adult, every cell can be visualized throughout the life cycle in living worms. Because a worm develops through an invariant cell lineage, knowledge of an imaged worm's lineage allows unambiguous assignment of the identity of each cell⁸. This potentially allows the alignment of expression patterns from individual worms onto a reference lineage, providing an integrated view of gene expression for each cell.

To exploit these possibilities systematically we had previously developed methods that allow automated computational tracing of the *C. elegans* lineage⁹⁻¹¹. These methods use custom software to identify nuclei and track them over time, in three-dimensional (3D) movies of worm embryos ubiquitously expressing nuclear-localized histone-GFP fusion proteins. This system generates highly accurate lineages through the 350-cell stage, which comprises all but the last round of embryonic cell division. Here we extend the system to provide detailed spatiotemporal characterization of reporter gene expression in both wild-type and genetically perturbed embryos.

RESULTS

Annotating reporter expression with the lineage

To monitor gene expression we generated promoter-histone gene-fluorescent protein gene reporter constructs, using a gene's 5' intergenic sequence to drive expression of an mCherry^{12,13} or a DsRed reporter (Table 1). Fusing the reporter to a *C. elegans* histone coding sequence directs the reporter to the nucleus to facilitate quantification and assignment of the signal to specific cells. These constructs, although lacking post-transcriptional control signals and perhaps even some transcriptional controls, should drive spatiotemporal expression of the reporter and provide a test of our ability to describe expression patterns over time in single cells. We used microparticle bombardment to generate stably integrated transgenes, to avoid high copy number, mosaicism,

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Table 1 | Strains and replicates

Promoter	Reporter	Strain	Number of embryos
<i>pha-4</i> (F38A6.1a ^a (4.1 kb))	HIS-24::mCherry	RW10062	4
<i>cnd-1</i> (3.2 kb)	HIS-24::mCherry	RW10055	2
<i>cnd-1</i> (3.2 kb)	HIS-24::mCherry	RW10060	2
<i>cnd-1</i> (3.2 kb)	HIS-24::mCherry	RW10083	1
<i>hlh-1</i> (3.3 kb)	HIS-24::mCherry	RW10097	2
<i>hlh-1</i> (3.3 kb)	HIS-24::mCherry	RW10112	1
<i>end-3</i> (1.0 kb)	HIS-24::mCherry	RW10064	2
<i>pha-4</i> (F38A6.1a ^a (4.1 kb))	HIS-24::DsRed	RW10007	6

^aTranscript used to define promoter boundary.

silencing and other artifacts associated with extrachromosomal arrays. We placed the resultant transgenes in a background ubiquitously expressing histone-GFP to follow the lineage. We adapted our previous imaging methods⁹ to collect two-color 3D movies from embryos by time-lapse confocal microscopy and traced the embryonic lineages and visualized the results with the programs StarryNite¹⁰ and AceTree¹¹ (**Supplementary Video 1** online).

We selected four developmentally important and well-studied transcription factors for initial testing (**Table 1**). *pha-4* encodes a FoxA transcription factor homologous to *Drosophila* forkhead required for specification of the pharynx^{14–16}. *cnd-1* encodes a NeuroD ortholog required throughout metazoans for proper neuron development¹⁷. *hlh-1* encodes a MyoD homolog important for muscle development¹⁸. Finally, *end-3* encodes a GATA-binding transcription factor important in specifying the intestine¹⁹. We used manual review of movies collected of reporter strains for each gene, to identify reporter-expressing cells through the 350-cell stage and compared our annotations to descriptions from the literature. We describe expression using the conventional *C. elegans* naming scheme (**Fig. 1**).

Our results generally agree with those in the literature (**Table 2**), and the minor differences are consistent with known differences in the specifics of the experiments. For example, for *end-3* the earliest

visible fluorescence appeared during the E2 stage (daughters of the E cell), but *end-3* transcript is detectable by *in situ* hybridization 15–30 min earlier in the late E1 stage¹⁹. This difference likely reflects the time required for translation and folding of the mCherry protein (15 min half-maturation time in *Escherichia coli*). Timing of fluorescence onset for the other reporters described here was also consistent with a 15–30 min lag. For *hlh-1*, we detected reporter signal not only in the myoblast lineages (D, Cap and Cpp), but also in the lineage MS, which gives rise to pharyngeal cells and glia as well as body wall muscle cells. Transient *hlh-1* expression had been previously detected in all descendants of the MS cell using protein-reporter fusions¹⁸; the more persistent expression we found here likely reflects the use of the promoter-reporter constructs and the stability of the histone-reporter fusion. For *pha-4*, the brighter expression we found in the E lineage relative to the pharyngeal precursors in the AB and MS lineages was similar to that of other reporter patterns and differed from antibody staining patterns of native protein^{15,16}, which reveal more abundant protein in the pharyngeal lineages. Presumably, these differences reflect regulatory sequences missing in the reporter constructs.

Our annotations extended the expression pattern analysis to cellular resolution and in so doing revealed new spatial and temporal details. Notably, for *cnd-1* we detected low signal consistently above background throughout the AB lineage from the 24-cell stage onward, with much brighter expression in a subset of AB cells starting at the 100-cell stage, as was reported previously¹⁷. However, the previous work did not specifically identify these brightly fluorescent cells other than to state that a few were likely to be ventral cord motor neurons and that most did not express UNC-86 protein. We identified the brightly fluorescent sublineages, which produce ring and other neurons in addition to motor neurons (**Table 2**). Our broader description of *cnd-1* promoter activity (114 neurons and glia versus 13 non-nervous system cells) suggests this gene may have a more complex role in neurogenesis than previously suggested.

Figure 1 | Lineage-based expression data display.

(a) Overview of the *C. elegans* lineage, displayed using the conventional naming scheme⁸. In this naming, initial divisions lead to a set of founder cells (AB, C, D, E, MS and P4), with subsequent daughters named based on their orientation at birth relative to the primary embryonic axes (anterior-posterior, dorsal-ventral and left-right). For example, ABal is the anterior daughter of ABal, which is itself the left daughter of ABa. Branch color indicates the predominant fate of terminal cells within each lineage (purple, neurons and neuronal support cells; yellow, pharynx; blue, epidermis; red, muscle; orange, intestine; gray, germline). (b) Embryonic lineage tree color-coded by *pha-4* reporter expression. Branches are ordered as in a. The pharynx is formed from most of the cells in the lineages ABalpa, ABaraa, MSaa and MSpa. E forms the intestine. Colors below the lineage show the encoding used in d. (c) 3D projection of a 350-cell stage RW10007 embryo with *pha-4* reporter expression (red) and ubiquitously expressed histone-GFP fusion proteins (yellow). (d) 3D model of the same embryo as in c generated from automated lineaging data. Reporter-expressing cells (>5 units) are color-coded by lineage identity, as labeled below the lineage tree in b. Non-reporter-expressing cells are semitransparent.

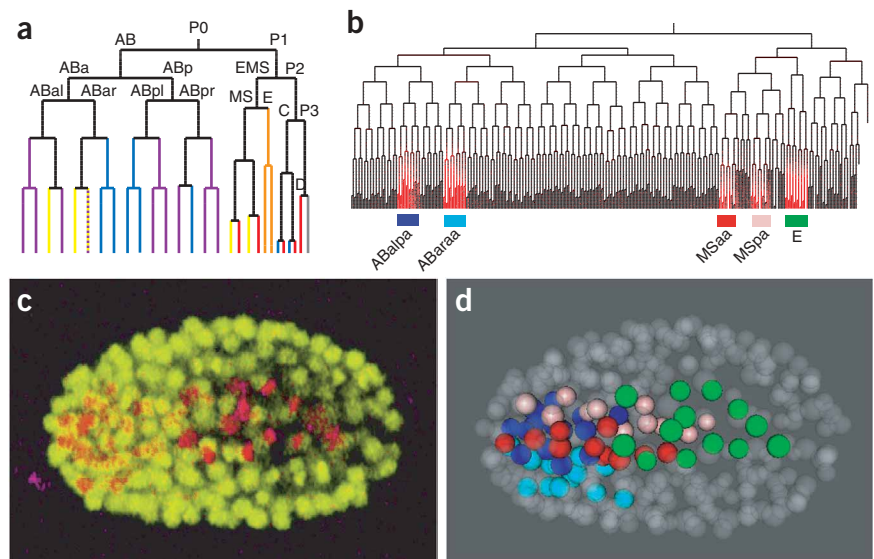


Table 2 | Comparing lineage-based expression patterns to descriptions from literature

Gene	Cells described in literature	Onset described in literature	Observed cells (all terminal fates in parentheses)	Observed onset
<i>pha-4</i>	Pharynx ¹⁴	> 500 cells	ABalpaxxx, ABaraaxxx, ABarapaxxx (57 pharyngeal cells, 4 neurons, 4 hypodermal cells)	< 200 cells
			MSaaxxx, MSpaxxx (37 pharyngeal cells, 2 neurons, 10 muscle cells)	< 200 cells
	Rectal precursors ¹⁴	> 500 cells	ABprppppxx, ABprppppax, ABplppppax, ABplppppxx (7 rectal and digestive muscle cells, 2 neurons, 1 muscle cell)	> 350 cells
<i>end-3</i>	Intestine (E) ¹⁴	100–200 cells	Exxx (20 intestine cells)	< 200 cells
	Intestine blast cell (E) ¹⁹	28 cells	Both E daughters (20 intestine cells)	2E (< 50 cells)
	<i>hlh-1</i> Muscle precursors and transiently in MS ¹⁸	12–24 cells (MS) 90+ cells (C)	MSxx (muscle and pharynx) Cpxx (Muscle) Dxx (Muscle)	24 cells (MS) 90 cells (C) 180 cells (D)
<i>cnd-1</i>	15 of 16 initial AB descendants ¹⁷	24 cells	Descendants of all 16 initial AB descendants with highly patterned expression	50–200 cells
	Non-hypodermal AB descendants not expressing UNC-86 (ref. 17)	By > 500 cells	ABalaaxx, ABalpaxx, ABalppax, ABarappx, ABplapax, ABplpax, ABplpax, ABpraax, ABprappx, ABprpax, ABprpax (97 neurons, 3 hypodermal cells, 1 arcade cell, 17 glia, 2 excretory system cells, 7 postembryonic blast cells)	100–200 cells

Automated description of expression

Because manual annotation of reporter-expressing cells is both subjective and time-consuming, we developed methods to automate quantitative measurements of these cells and to systematically and rigorously evaluate reproducibility within and between independently derived strains.

We calculated the mCherry fluorescence intensity within each nucleus at each time point (collected once per minute) using the nuclear positions and diameters estimated by StarryNite (roughly 20,000 measurements per embryo). Because in some cases out-of-focus light from nearby reporter-expressing nuclei can increase signal within non-reporter-expressing nuclei, leading to false positives, we subtracted locally calculated background from each expression measurement. The resulting high-resolution expression data can be displayed as a color-coded lineage tree or in a spatial representation of the lineage (Fig. 1 and Supplementary Video 2 online). To distinguish non-expressing cells from reporter-expressing cells we developed methods to estimate the statistical significance and time of onset of expression for each cell and also estimated the time of onset of expression for the lineage of each terminal cell in each recording (Supplementary Data online).

The measured intensity values varied continuously from near 0 (-0.07 ± 0.69 (arbitrary) units per pixel) for non-reporter-expressing cells in negative control embryos to very high values (> 100) for cells expressing high levels of reporter. Even among reporter-expressing cells in a single embryo we observed a large dynamic range. For example, peak expression of a *cnd-1* reporter in the mostly neuron-producing ABarapp lineage were 58–116 units ($P < 10^{-22}$), whereas that in its pharynx-producing sister lineage ABarapa was also significant ($P < 10^{-19}$) but reached only 6–7 units.

Analysis of two negative control embryos with no mCherry transgene gave no cells with significant expression, with a cutoff of $P < 3 \times 10^{-6}$ (Supplementary Fig. 1 online). Using the same threshold, cells in reporter-containing embryos with significant

fluorescence corresponded well to the cells identified as reporter-expressing by manual inspection. Several sublineages not annotated manually also passed this cutoff. A few were likely false positives, but others represent previously overlooked expression. For example, all three *cnd-1* reporter strains had significant expression throughout the EMS lineage, which generates mostly endoderm and mesoderm. This expression was dimmer than that in the AB cells (2–10 units in EMS cells compared with 10–100 units for the AB cells), which may be why it was not identified in previous studies.

Automated estimates of the time of expression onset agreed with the manual review of images as well as with inspection of the plots of brightness versus time for each terminal cell. For example, the automated estimate of the time of *end-3* reporter expression onset in the E lineage was in the E2 stage, the same as was annotated manually.

Reproducibility of expression patterns

We compared replicate image series for strains expressing reporter constructs for each of the four genes to assess the reproducibility of the measured expression, of the time of expression onset and of the identities of significantly fluorescent cells in the reporter strains using our automated methods.

Relative reporter expression for each cell (normalized to that in other cells in the same embryo) was reproducible from one series or strain to the next, despite the many possible sources of variation (Supplementary Figs. 2–4 and Supplementary Table 1 online). Comparing the average intensity of each cell across four embryos of a *pha-4* reporter strain yielded high reproducibility (mean correlation coefficient across all comparisons ($r = 0.96$) as did comparing replicate expression patterns for *cnd-1* ($r = 0.92$), *hlh-1* ($r = 0.94$) and *end-3* ($r = 0.95$) reporter strains. As additional variability might be introduced by the strain construction process, we compared two independent *hlh-1* reporter strains made with the same construct; these measurements were also highly correlated (mean

$r = 0.95$) as were three independent strains containing the same *cnd-1* reporter construct (mean $r = 0.86$).

In contrast, a second *pha-4* reporter strain made with a different fluorescent reporter (unoptimized DsRed.T1 instead of worm-optimized mCherry) and transformation marker (*pha-1* instead of *unc-119*) had more variable expression (mean $r = 0.84$; range 0.68–0.95). Examining the lineage trees and expression for this strain (**Supplementary Fig. 3**) identified substantial differences in the identities of reporter-expressing cells: some had bright expression in the ABalpp, ABarp and MSxp lineages. These lineages give rise predominantly to nonpharyngeal cells and are the sister lineages of predominantly pharyngeal lineages that express the *pha-4* reporter in all embryos. Whereas determining the cause and biological relevance of this observation will require additional experiments, our ability to detect the differences in pattern demonstrates the utility of lineage analysis in identifying and characterizing variability in reporter expression.

The identities of reporter-expressing cells were also highly reproducible between embryos of the same strain. Of highly significantly fluorescent cells ($P < 10^{-12}$), 94.3% (8,192/8,683) were significant at $P < 10^{-6}$ in a second embryo containing the same reporter construct (38 comparisons involving 14 embryos for four genes). Even of cells with intermediate expression significance ($10^{-12} < P < 10^{-6}$), in 68%, expression was significant in the second embryo (1,054/1,550; **Supplementary Fig. 5** online). It is not surprising that the confirmation rate was lower for these cells because they have much lower expression (average, 3.3 units) than cells in the more significant expression set (average, 25 units) and were thus closer to the detection limit of the system. Supporting this, the confirmation rate increased to $> 97\%$ if we considered only cells reaching an arbitrary threshold of 10 units, and over half of all unconfirmed cells had significant expression at the $P < 0.01$ cutoff in the second embryo (compared with 10% of cells in the negative control embryos).

The majority of cells with expression brighter than 5 units that we could not confirm in the second series represented real differences in expression between embryos based on manual inspection (13 of 15 cases tested). For cells below this cutoff, 34% (113 of 333 tested) represented clear false positives in one series, caused by imaging artifacts such as dust flecks and coverslip reflections. The remaining cells were difficult to score: cells missed at this level could be either not expressing the reporter or expressing below our sensitivity cutoff. The relatively small number of differences suggests that it should be possible to reliably identify all brightly fluorescent cells with 2–3 replicate image series, and 4–5 or more replicates would allow higher sensitivity for dimly or occasionally fluorescent cells.

The estimated onset of expression for the four genes ranged from 48 min to 154 min after the ABa (4-cell stage) division. Across all reporter-expressing cells, the median s.d. of onset time in replicate

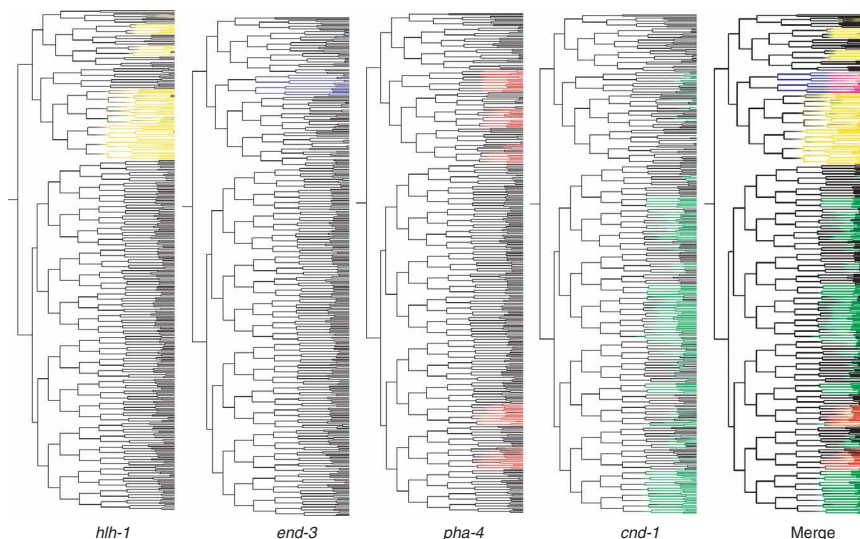


Figure 2 | Comparison of expression patterns. Expression-coded lineage trees for strains RW10055 (*cnd-1*(3.2 kb)::HIS-24::mCherry) (green), RW10064(*end-3*(1.0 kb)::HIS-24::mCherry) (blue), RW10007(*pha-4*(4.1 kb)::HIS-24::DsRed) (red) and RW10097(*hlh-1*(3.3 kb)::HIS-24::mCherry) (yellow) reporters. Branch order is the same as in **Figure 1**.

embryos was 11 min and 90% of cells had an s.d. of less than 20 min (**Supplementary Fig. 6** online). By comparison, the mean cell cycle length through the 200-cell stage was 32 min.

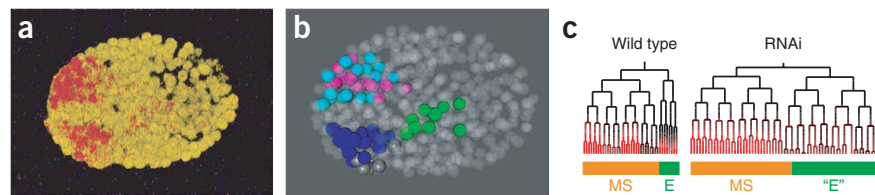
Integrating expression patterns

The invariance of the wild-type lineage allows mapping of multiple expression patterns onto a single reference lineage tree to examine their relationships. For example, the expression patterns of the *pha-4*, *cnd-1* and *hlh-1* reporters are largely orthogonal, but the *end-3* reporter is coexpressed in the E lineage with the *pha-4* reporter and has an earlier onset (**Fig. 2**). These relationships can be expressed in terms of correlation coefficients: the mean correlation for *pha-4* and *end-3* reporters was 0.68 compared with -0.002 for all other inter-gene comparisons. As a result, relationships between expression patterns can be systematically explored by hierarchical clustering (**Supplementary Fig. 7** online). Expression patterns of replicate embryos formed tight clusters, and the relationship between reporters with similar expression, notably *end-3* and *pha-4*, was evident in the clustering patterns, as were the differences in expression in embryos expressing *pha-4*::HIS-24::DsRed. With a larger dataset and more sophisticated comparison algorithms, this type of analysis may be useful in identifying the modules that make up the embryonic transcriptional regulatory network.

Detecting quantitative changes in reporter expression

Although expression patterns were well correlated between embryos expressing the same reporter, we observed variability in absolute expression (acting multiplicatively on all cells of a given embryo) over a twofold range. This is not surprising owing to the many technical and biological factors that can influence the reporter signal. We reasoned, however, that with sufficient replicates and controls it might be possible to use the magnitude of expression as a quantitative readout of developmental pathways.

Figure 3 | Expression changes after RNAi. (a) A 3D projection of an embryo expressing *pha-4*–*DsRed.T1* (red, DsRed; yellow, ubiquitously expressed histone-GFP fusion) whose mother was fed bacteria expressing *lit-1* double-stranded RNA (RW10007). Note multiple isolated regions of expression while the wild type (Fig. 1c) has a single cohesive expression domain. (b) A 3D model of the embryo in a with reporter-expressing cells colored as in Figure 1d with the addition of the nonpharyngeal lineages ABarap (gray) and ABalpp (magenta). (c) EMS sublineage tree with *pha-4* reporter expression in embryos from wild-type and *lit-1* (RNAi) mothers, showing extra cells and reduced *pha-4* expression in *lit-1* E lineage.



We tested the effect of the GATA transcription factor encoded by the gene *elt-7* on the *pha-4*::HIS-24::mCherry reporter in the E lineage. ELT-7 is thought to regulate early gut differentiation redundantly with the GATA factor encoded by *elt-2* (ref. 20). The *pha-4* promoter is known to contain sites bound *in vitro* by ELT-2 and ectopic expression of ELT-2 can lead to ectopic expression of *pha-4* reporters¹⁵. However, the effect of ELT-7 on *pha-4* reporter expression is unknown. We compared *pha-4* reporter intensity in the gut cells of comma-stage wild-type embryos ($n = 28$) with intensity in embryos homozygous for a deletion allele of *elt-7* ($n = 56$). With these large numbers and a careful protocol (see **Methods**), we observed a mean 23% decrease in *pha-4* reporter intensity ($P = 3.9 \times 10^{-7}$). This difference cannot be explained by differences in the z position of the reporter-expressing cells. We noted a similar effect when we targeted *elt-7* by RNA interference (RNAi; data not shown), indicating that genetic background differences are not responsible for the effect. These data suggest that ELT-7 is required for full *pha-4* reporter expression in the E lineage. The residual expression may be due to activation by ELT-2, for which inactivation by RNAi shows a similar effect (data not shown), and possibly by other factors acting at this promoter.

Characterizing expression in altered lineages

Our quantitative image analysis methods can also detect changes in both expression pattern and expression level at cellular resolution in altered lineages. Some of the cells that express the *pha-4* reporters derive from the E and MS founder cells. E and MS are sister cells with different fates because of a WNT signal received by E but not MS^{21,22}. Loss of function of the gene *lit-1*, which encodes a NEMO-like kinase in the WNT pathway, causes E to adopt the MS fate²³. Disrupting *lit-1* function by RNAi leads to a change in the spatial expression of the *pha-4* reporter (Fig. 3). The embryo does not gastrulate, altering its anatomy dramatically and making it difficult to determine the identity of the reporter-expressing cells anatomically. The lineage patterns combined with the *pha-4* expression pattern, however, reveal that the central misplaced reporter-expressing cells are generated from the E founder cell, which has adopted an MS-like fate. Some of the reporter-expressing cells in the anterior of the embryo have the same lineage identity (descended from ABalpa, ABaraa) as reporter-expressing cells in wild-type embryos, and are simply misplaced because of gastrulation defects. However, many cells (most cells in the ABalpp and ABarap lineages) show robust ectopic expression not seen in wild type. These cells normally give rise to non-pharyngeal cells and these data suggest that WNT signaling acts to prevent them from adopting a pharyngeal fate and expressing *pha-4*.

DISCUSSION

The methods described here allow integration of expression patterns from different genes in different worms, providing the potential to obtain a comprehensive picture of gene expression in every cell. For example, using this methodology it should be possible to describe the patterns of activity of the promoters for all of the transcription factor genes active during embryogenesis. Combined with emerging knowledge of transcription-factor binding sites, these expression patterns would begin to reveal the network of regulatory control. Predicted networks could then be tested by quantitative analysis of reporters after RNAi or genetic depletion of predicted regulators. With appropriate controls for integration site and copy number, promoter dissection could allow the identification of the specific DNA sequences required for each regulator's effects.

The ability to trace altered lineages and generate a quantitative readout of cell type-specific reporters extends the phenotypic analysis of mutants and RNAi treatments that perturb development. As the number of cell fate markers increases, the method will become increasingly powerful in phenotypic analysis that will be required to gain a functional understanding of regulatory networks.

Future developments could considerably enhance the already powerful information obtained. Extending automated lineaging to include the last round of embryonic cell division will facilitate describing the expression of those genes only expressed in the terminally differentiated cells. This remains a challenging goal because of the active cell migrations and close packing of nuclei in the late embryo. Another improvement would be the development of a red fluorescence-based lineaging system, which would allow the embryonic expression patterns for existing genomically integrated GFP reporters^{24–26} to be characterized. Whether the many other non-integrated strains available would be useful for systematic analysis is questionable because their mosaicism would be expected to necessitate additional replicates to ensure identification of all reporter-expressing cells and because a fraction of embryos imaged would not contain the transgene at all.

Our results emphasize that promoter-reporter constructs only partially capture the complexity of regulation and that there is the need for more faithful transgenic strategies. We intentionally used transcriptional fusions to histone genes for this study to maximize sensitivity for weakly expressed genes. Protein fusions could also be generated to reveal subcellular localization of the proteins and to contrast transcriptional and translational controls, at the potential expense of reduced sensitivity for proteins with faster turnover than HIS-24. Improved cloning and transformation methods such as recombineering²⁷ would allow the use of larger genomic segments so that the observed patterns are likely to more faithfully

reflect the native pattern. Faster-folding, brighter reporters would ensure that the system detects the earliest expression of even weakly expressed genes. Extension of the system to multiple colors could increase throughput and provide kinetic information about colocalization.

In vivo single-cell analysis of gene expression is an important step toward a comprehensive molecular understanding of development. The transparency of the worm and its invariant lineage make it ideal for such analyses, but with continued progress in noninvasive imaging to track cells and ever-expanding sets of cell-fate markers to substitute for the lineage, we envision equivalent analyses for more complex organisms, including mammals.

METHODS

Imaging. We performed confocal imaging with a Zeiss LSM510 as described⁹ with the addition of parallel acquisition of mCherry (or DsRed) signal (see **Supplementary Methods** online). We collected time points once per minute between the four-cell and comma stages (5–6 h). We used the same mCherry or DsRed imaging settings throughout each recording. All expression patterns reported are from worms whose development proceeded normally. When scored, these worms hatched into L1 larvae with normal morphology, and the lineages were identical to those in the wild type through the 350-cell stage. Although fluctuations in laser output over time would be predicted to alter the absolute magnitude of expression in an embryo, we measured this output regularly (about once per month) during the project and found that intensity varied within a modest range (<10%).

Lineage analysis. We generated and edited lineages with the programs StarryNite^{9,10} and AceTree¹¹ as described.

Quantitative intensity analysis. We compared three different strategies to quantify the red fluorescence signal. In each case, we began by summing the signal within each nucleus. For one strategy, we did not subtract the background. We observed that on occasion this led to signal from strongly fluorescent cells interfering with the measurements of nearby cells, resulting in false positives. To address this problem, we tested two strategies in which we subtracted locally computed background signal either from the raw images (unmasked) or from images in which nearby cells were excluded from the background calculation (masked; see **Supplementary Methods** and **Supplementary Table 2** online), and chose the more conservative ‘unmasked’ strategy.

To generate the expression intensity values along the lineage, we defined the local background, $B_{n,t}$, for each nucleus (n) in the red channel images at each time point (t) as the average pixel intensity of pixels between $1.2r$ and $2r$ from the nuclear centroid, where r is the radius of the nucleus. We defined the raw intensity, $R_{n,t}$, as the average intensity of pixels within the bounds of the predicted nucleus. We calculated the corrected intensity, $I_{n,t}$, as $R_{n,t} - B_{n,t}$ and used this for all subsequent analysis. We implemented this algorithm (available upon request) as a stand-alone extension to AceTree. We visualized expression-coded trees and projections using AceTree¹¹.

An important imaging issue that could obscure subtle expression differences between lineages is depth; typically intensity decreases with depth in the specimen. The variable excitation light settings with depth were designed to reduce this effect (see

Supplementary Methods for a further discussion of the residual impact of depth on quantification).

For hierarchical clustering, we first aligned the branches of each lineage tree to a reference lineage (see **Supplementary Methods**). We linearized the aligned data into a single vector for each series, clustered using Cluster (<http://fafner.stanford.edu/~sherlock/cluster.html>), and visualized them in Java TreeView²⁸.

For the *elt-7* expression analysis, we collected single z stacks for wild type ($n = 28$) or *elt-7(ok835)* ($n = 56$) worms using the same image settings as we used for image series collection. We acquired the images in 3 sessions; in each one roughly comparable numbers of wild type and mutant embryos were imaged to ensure variation in laser power over time or other factors were not confounded with mutant status. We used StarryNite to assign nuclei based on mCherry or DsRed expression; nuclei with locations and sizes compatible to E-derived nuclei were selected using a script, and the average raw mCherry or DsRed expression per nucleus was calculated for each embryo. The observed differences cannot be explained by differences in z position of reporter-expressing cells: we measured the effect of this factor to be approximately 3% per plane (see **Supplementary Methods**). The positions of the E cells in wild-type and *elt-7* embryos were very similar, with the positions in the wild type being lower by 1.5 planes on average. Given that our analysis suggests lower z plane causes reduced expression, the actual expression difference is likely slightly larger than was measured.

Statistical analysis. We evaluated the significance of the reporter expression in each terminal cell (cells that did not divide during the recording) as follows. First, we calculated the trajectory of expression values of each cell and its parents, back to the first time point for that embryo. For each time point in each trajectory, we calculated the significance for expression beyond that time point being greater than local background by using a Wilcoxon signed rank sum test. This led to a time series of P values for each cell trajectory. For reporter-expressing cells, a global minimum could be identified: in 10/10 cases examined in detail, the last time point that P was within 100-fold of this minimum was within five time points of the point subjectively identified as the onset of fluorescence. This method was more robust for weakly expressing genes than choosing an arbitrary intensity threshold as the onset time. Examining the significance levels determined this way in two image series with no RFP transgene identified $P < 10^{-6}$ as a threshold that would lead to less than one false positive per series. We compared the distributions of red fluorescence per embryo in wild-type and *elt-7* mutants by a Wilcoxon rank sum test with continuity correction.

Quantitative comparisons of replicates. Although we generated multiple strains for both the *cnd-1* and *hlh-1* reporters, we grouped these strains together for the analysis of reproducibility because expression was not substantially more different between strains than it was between embryos of the same strain. To identify the rates at which cells significant at different levels were confirmed in a second series, we generated a list of all cells for all 38 pairwise combinations of replicate embryos containing the same reporter construct and filtered them based on various criteria. To reduce the effect of outliers, peak expression level was defined as the second highest measured expression value in a cell’s history. We then assessed the fraction of cells that would be called as

expressing with the cutoff $P < 0.01$ in the second embryo. For comparison, $<10\%$ of cells in the negative control embryos would be called as expressing with the cutoff $P < 0.01$. To describe variability in time of onset, we identified terminal cells identified as expressing ($P < 10^{-6}$) in each replicate for a given gene, then calculated the s.d. of the time of onset for each cell. Because the onset was usually several cell cycles before the terminal stage, related terminal cells frequently share a common point of onset. To avoid overcounting these cells, we limited subsequent analysis to unique expression onset events. We quantitatively compared expression by first calculating the average expression level for each cell and then comparing these average expression levels between two embryos.

Additional methods. Descriptions of constructs, strain construction, RNAi, microscope settings, background subtraction methods, tree alignment, the effect of depth on quantitation, and image processing are available in **Supplementary Methods**.

Note: Supplementary information is available on the Nature Methods website.

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AUTHOR CONTRIBUTIONS

R.H.W., Z.B. and J.I.M. conceived the experiments. J.I.M., Z.B., M.E.B., T.J.N., B.L.M., Z.Z. and M.J.S. performed the experiments. T.J.B., Z.B. and J.I.M. wrote software for data analysis. J.I.M., Z.B., T.J.B. and J.I.M. analyzed the data. J.I.M. and R.H.W. wrote the paper with input from the other authors.

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- Lawrence, P.A., Johnston, P., Macdonald, P. & Struhl, G. Borders of parasegments in *Drosophila* embryos are delimited by the fushi tarazu and even-skipped genes. *Nature* **328**, 440–442 (1987).
- Fox, R.M. *et al.* A gene expression fingerprint of *C. elegans* embryonic motor neurons. *BMC Genomics* **6**, 42 (2005).
- Baugh, L.R., Hill, A.A., Slonim, D.K., Brown, E.L. & Hunter, C.P. Composition and dynamics of the *Caenorhabditis elegans* early embryonic transcriptome. *Development* **130**, 889–900 (2003).
- Baugh, L.R. *et al.* The homeodomain protein PAL-1 specifies a lineage-specific regulatory network in the *C. elegans* embryo. *Development* **132**, 1843–1854 (2005).
- Furlong, E.E., Andersen, E.C., Null, B., White, K.P. & Scott, M.P. Patterns of gene expression during *Drosophila* mesoderm development. *Science* **293**, 1629–1633 (2001).
- Arbeitman, M.N. *et al.* Gene expression during the life cycle of *Drosophila melanogaster*. *Science* **297**, 2270–2275 (2002).
- Zhang, Y. *et al.* Identification of genes expressed in *C. elegans* touch receptor neurons. *Nature* **418**, 331–335 (2002).
- Sulston, J.E., Schierenberg, E., White, J.G. & Thomson, J.N. The embryonic cell lineage of the nematode *Caenorhabditis elegans*. *Dev. Biol.* **100**, 64–119 (1983).
- Murray, J.I., Bao, Z., Boyle, T. & Waterston, R.H. The lineaging of fluorescently-labeled *Caenorhabditis elegans* embryos with StarryNite and AceTree. *Nat. Protoc.* **1**, 1468–1476 (2006).
- Bao, Z. *et al.* Automated cell lineage tracing in *Caenorhabditis elegans*. *Proc. Natl. Acad. Sci. USA* **103**, 2707–2712 (2006).
- Boyle, T.J., Bao, Z., Murray, J.I., Araya, C.L. & Waterston, R.H. AceTree: a tool for visual analysis of *Caenorhabditis elegans* embryogenesis. *BMC Bioinformatics* **7**, 275 (2006).
- McNally, K., Audhya, A., Oegema, K. & McNally, F.J. Katanin controls mitotic and meiotic spindle length. *J. Cell Biol.* **175**, 881–891 (2006).
- Shaner, N.C. *et al.* Improved monomeric red, orange and yellow fluorescent proteins derived from *Discosoma* sp. red fluorescent protein. *Nat. Biotechnol.* **22**, 1567–1572 (2004).
- Azzarà, M., Goszczynski, B., Chung, M.A., Kalb, J.M. & McGhee, J.D. A fork head/HNF-3 homolog expressed in the pharynx and intestine of the *Caenorhabditis elegans* embryo. *Dev. Biol.* **178**, 289–303 (1996).
- Kalb, J.M. *et al.* pha-4 is Ce-fkh-1, a fork head/HNF-3alpha,beta,gamma homolog that functions in organogenesis of the *C. elegans* pharynx. *Development* **125**, 2171–2180 (1998).
- Horner, M.A. *et al.* pha-4, an HNF-3 homolog, specifies pharyngeal organ identity in *Caenorhabditis elegans*. *Genes Dev.* **12**, 1947–1952 (1998).
- Hallam, S., Singer, E., Waring, D. & Jin, Y. The *C. elegans* NeuroD homolog cnd-1 functions in multiple aspects of motor neuron fate specification. *Development* **127**, 4239–4252 (2000).
- Krause, M. MyoD and myogenesis in *C. elegans*. *Bioessays* **17**, 219–228 (1995).
- Maduro, M.F. *et al.* Genetic redundancy in endoderm specification within the genus *Caenorhabditis*. *Dev. Biol.* **284**, 509–522 (2005).
- Maduro, M.F. & Rothman, J.H. Making worm guts: the gene regulatory network of the *Caenorhabditis elegans* endoderm. *Dev. Biol.* **246**, 68–85 (2002).
- Lin, R., Hill, R.J. & Priess, J.R. POP-1 and anterior-posterior fate decisions in *C. elegans* embryos. *Cell* **92**, 229–239 (1998).
- Lin, R., Thompson, S. & Priess, J.R. pop-1 encodes an HMG box protein required for the specification of a mesoderm precursor in early *C. elegans* embryos. *Cell* **83**, 599–609 (1995).
- Kaletta, T., Schnabel, H. & Schnabel, R. Binary specification of the embryonic lineage in *Caenorhabditis elegans*. *Nature* **390**, 294–298 (1997).
- Bieri, T. *et al.* WormBase: new content and better access. *Nucleic Acids Res.* **35**, D506–D510 (2007).
- Hunt-Newbury, R. *et al.* High-throughput *in vivo* analysis of gene expression in *Caenorhabditis elegans*. *PLoS Biol.* **5**, e237 (2007).
- Reece-Hoyes, J.S. *et al.* Insight into transcription factor gene duplication from *Caenorhabditis elegans* promoterome-driven expression patterns. *BMC Genomics* **8**, 27 (2007).
- Sarov, M. *et al.* A recombineering pipeline for functional genomics applied to *Caenorhabditis elegans*. *Nat. Methods* **3**, 839–844 (2006).
- Saldanha, A.J. Java Treeview—extensible visualization of microarray data. *Bioinformatics* **20**, 3246–3248 (2004).